Annex G. Glossary

María de los Ángeles Olaya García; Jhon Jairo Cortez Vargas

Servicio Nacional de Aprendizaje (SENA), Programa ADSO-3145556

Instructora: Karol Daniela Correa

12 de octubre de 2026

Annex G. Glossary

• | Term | Definition |

• |---|---|

• | ADR | Architecture decision record: a short document explaining a design decision and its consequences |

• | API | Interface that enables communication between applications |

• | API Gateway | Single entry point that routes requests, validates the JWT and limits traffic |

• | BFF | Backend-for-Frontend: a server layer tailored to the needs of one client |

• | BLE | Bluetooth Low Energy; the mobile app uses it to provision the ESP32 |

• | DHT11 | Digital sensor for air temperature and humidity |

• | ESP32 | Microcontroller with Wi-Fi and Bluetooth |

• | Hexagonal (architecture) | Style that separates the domain from inbound and outbound adapters through ports |

• | IoT | Internet of Things: connected physical devices that send data |

• | JWT | Signed token that authenticates a user's requests |

• | JWKS | Set of public keys that services use to verify JWT signatures |

• | KY-038 | Sound module with analog output for a relative noise level |

• | Liquibase | Tool for versioning database changes |

• | MQ-135 | Gas and air-quality sensor; requires calibration and does not measure CO■ in ppm without it |

• | MQTT | IoT messaging protocol considered in the architecture; not confirmed in the firmware |

• | OAuth2 / OIDC | Authorization and identity standards used for sign-in |

• | OpenAPI | Standard for describing REST API contracts |

• | Outbox | Pattern that stores events in the database before publishing them so they are not lost |

• | RBAC | Role-based access control (SUPER_ADMIN, ADMIN, USER, VIEWER) |

• | Smoke test | Quick end-to-end test confirming that the basic system responds |

• | Testcontainers | Library that starts Docker containers (for example PostgreSQL) during tests |

• | X-API-Key | Key an IoT device uses to authenticate when sending measurements |